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|--------------------|--|---------------------|--|
| <b>Student ID:</b> |  | <b>Lab Section:</b> |  |
| <b>Name:</b>       |  | <b>Lab Group:</b>   |  |

## Experiment No. 3

# Verification of Superposition Principle

### Objective

This experiment aims to experimentally verify the Superposition theorem, which is an analytical technique for determining currents/voltages in a circuit with more than one emf source.

### Theory

The Superposition Principle is a fundamental concept in electrical circuits that states that in any linear, active, bilateral network having more than one source, the response across any element is the sum of the responses obtained from each source considered separately, and all other sources are replaced by their internal resistance. The superposition theorem is used to solve networks where two or more sources are present and connected. The current or voltage through any component in a circuit is the sum of the effects of each individual source acting alone. In other words, the principle states that **the total response of a circuit with multiple sources is the sum of the responses of the circuit to each individual source acting alone.** This principle is widely used in circuit analysis to simplify complex circuits and solve them with ease.

In a **linear circuit** containing multiple independent sources and linear elements (e.g., resistors, inductors, and capacitors), the voltage across (or the current through) any element when all the sources are acting simultaneously may be obtained by adding algebraically all the individual voltages (or currents) caused by each independent source acting alone, with all other sources deactivated.

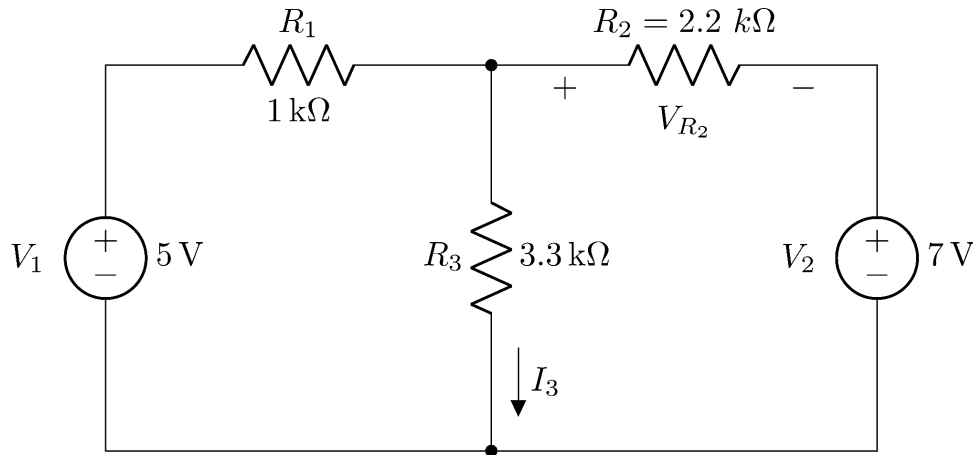
An independent **voltage source** is deactivated (made zero) by **shorting** it, and an independent **current source** is deactivated (made zero) by **open-circuiting** it. However, if a dependent source is present, it must remain active during the superposition process.

### Apparatus

- Multimeter
- Resistors (1 k $\Omega$ , 2.2 k $\Omega$ , 3.3 k $\Omega$ ).
- DC power supply
- Breadboard
- Jumper wires

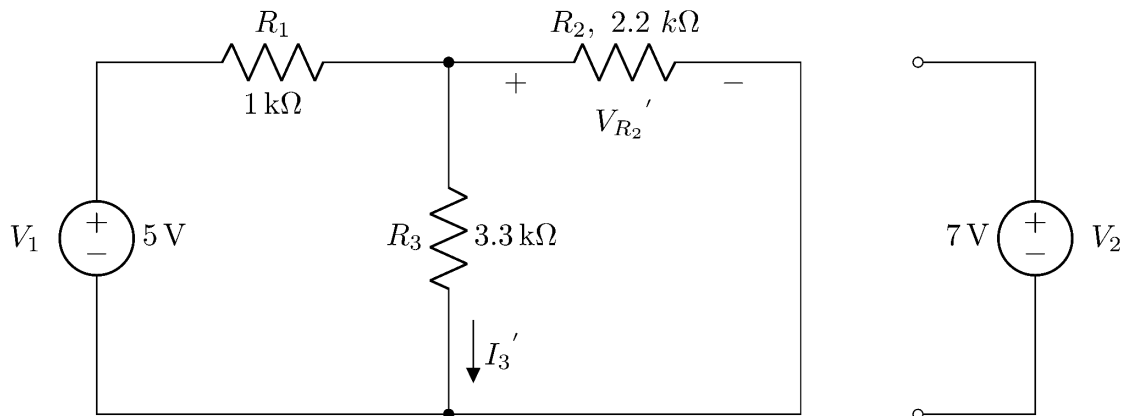
## Procedures

- Measure the resistances of the provided resistors and fill up the data table (Table 1).
- Construct the following circuit on a breadboard. Try to use minimum number of jumper wires:



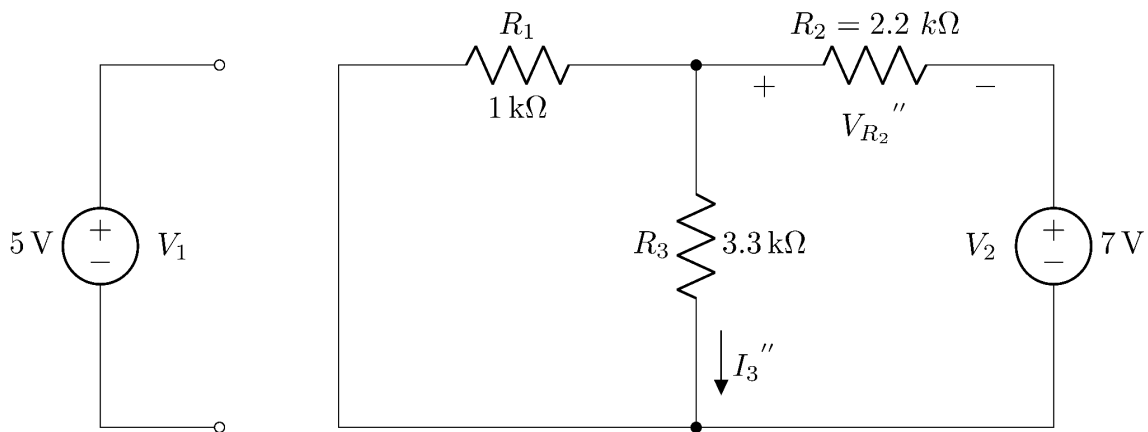
**Circuit 1**

- Measure the voltage across the resistors  $R_2$ ,  $R_3$  and current through the resistor  $R_3$ . Use a Multimeter to measure the voltage and use Ohm's law to calculate the current ( $I_3$ ) through  $R_3$ . Fill up the data tables.
- Render  $V_1$  inactive (keeping  $V_2$  active) and construct the following circuit.



**Circuit 2**

- Measure the voltage across the resistors  $R_2$ ,  $R_3$  and current through the resistor  $R_3$ . Use a Multimeter to measure the voltage and use Ohm's law to calculate the current ( $I_3'$ ) through  $R_3$ . Fill up the data tables.
- Render  $V_2$  inactive (keeping  $V_1$  active) and construct the following circuit.



**Circuit 3**

- Measure the voltage across the resistors  $R_2$ ,  $R_3$  and current through the resistor  $R_3$ . Use a Multimeter to measure the voltage and use Ohm's law to calculate the current ( $I_3''$ ) through  $R_3$ . Fill up the data tables.
- Verify if  $I_3 = I_3' + I_3''$ . This validates the superposition theorem for the current through  $R_3$ .
- Verify if  $V_{R_2} = V_{R_2}' + V_{R_2}''$ . This validates the superposition theorem for the voltage across  $R_2$ .

### Data Tables

Signature of Lab Faculty:

Date:

**\*\* For all the data tables, take data up to three decimal places, round to two, then enter into the table.**

### Table 0: Resistance Data

For all your future calculations, please use the observed values only (**even for theoretical calculations**).

| Notation | Expected Resistance | Observed Resistance (kΩ) |
|----------|---------------------|--------------------------|
| $R_1$    | 1 kΩ                |                          |
| $R_2$    | 2.2 kΩ              |                          |
| $R_3$    | 3.3 kΩ              |                          |

**Table 1: Current through  $R_3$  and voltage across  $R_2$**

In the following table,  $I_3$  is the current through  $R_3$  and  $V_{R_2}$  is the voltage drop across the resistor  $R_2$ . A similar syntax applies to the remaining resistors. Also, calculate the percentage of error between expected and observed values of  $I_3' + I_3''$ .

| Observation  | $I_3$ with both $V_1$ and $V_2$ active (mA)    | $I_3'$ with only $V_1$ is active (mA)    | $I_3''$ with only $V_2$ is active (mA)    | $I_3' + I_3''$ (mA)        |
|--------------|------------------------------------------------|------------------------------------------|-------------------------------------------|----------------------------|
| Experimental |                                                |                                          |                                           |                            |
| Theoretical  |                                                |                                          |                                           |                            |
| Observation  | $V_{R_2}$ with both $V_1$ and $V_2$ active (V) | $V_{R_2}'$ with only $V_1$ is active (V) | $V_{R_2}''$ with only $V_2$ is active (V) | $V_{R_2}' + V_{R_2}''$ (V) |
| Experimental |                                                |                                          |                                           |                            |
| Theoretical  |                                                |                                          |                                           |                            |

● **Percentage of error** =  $\left| \frac{\text{Observed Value} - \text{Expected Value}}{\text{Expected Value}} \right| \times 100\%$

*N.B: Here, the Expected values are  $I_3$ ,  $V_{R_2}$  and the Observed values are  $I_3' + I_3''$  and  $V_{R_2}' + V_{R_2}''$  respectively.*

Hence, Percentage of error in  $I_3' + I_3''$  calculation =

%

Hence, Percentage of error in  $V_{R_2}' + V_{R_2}''$  calculation =

%

## Questions

1. Refer to the Data Table 2 to answer the following questions—

(a) Calculate the **power associated** with  $R_2$  using the experimentally measured values of currents or voltages in Table 2 when:

- Only  $V_1$  source is active.
- Only  $V_2$  source is active.
- Both  $V_1$  and  $V_2$  sources are active.

Fill out the Table given below and verify, whether the superposition theorem is verified or not in this case. **You don't need to take any new readings for this task.**

Use previous data from Table 2 to calculate the power. Remember, power consumed by a resistor can be written as,  $P = VI = I^2R = \frac{V^2}{R}$

| Observation  | $P_{R_2}$ when both $V_1$ and $V_2$ are active<br>$P_{R_2} = \frac{V_{R_2}^2}{R_2}$<br>(mW) | $P_{R_2}'$ when only $V_1$ is active<br>$P_{R_2}' = \frac{V_{R_2}^{'2}}{R_2}$<br>(mW) | $P_{R_2}''$ when only $V_2$ is active<br>$P_{R_2}'' = \frac{V_{R_2}^{''2}}{R_2}$<br>(mW) | $P_{R_2}' + P_{R_2}''$<br>(mW) |
|--------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------|
| Experimental |                                                                                             |                                                                                       |                                                                                          |                                |
| Theoretical  |                                                                                             |                                                                                       |                                                                                          |                                |

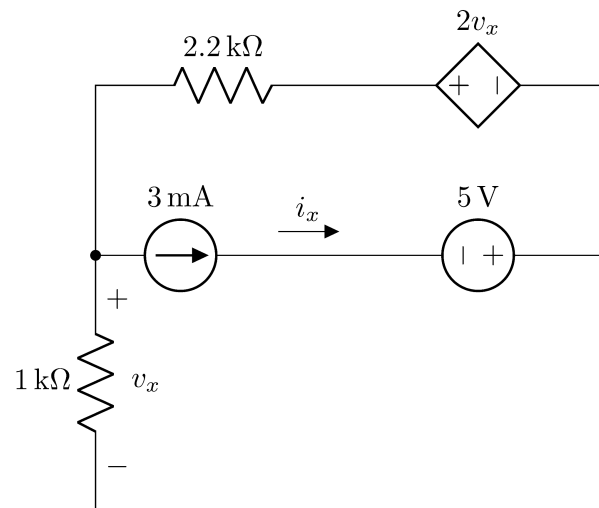
(b) Is the Superposition Principle applicable in the case of Power?

☐ Yes      ☐ No

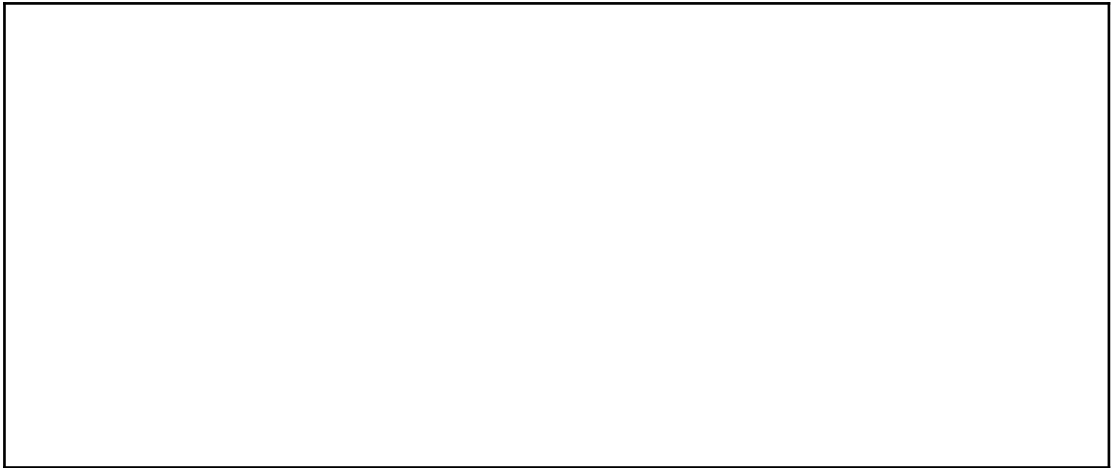
(c) How would you relate your findings from this to the concept of linearity? Why does/doesn't it work when it comes to Power?

2. Why was a short circuit wire required to be connected between the corresponding terminals while turning off a voltage source? Why wasn't simply turning off the power switch enough to deactivate the source?

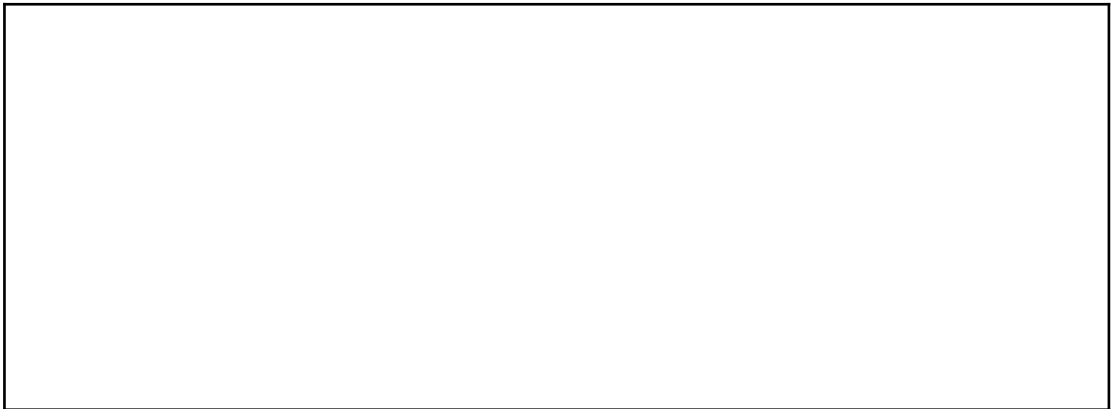
3. For the circuit shown below,



- (a) Show mathematically using the Superposition Principle that the 5 V voltage source in series with the current source has no contribution to the current  $i_x$ . [Hint: dependent or controlled sources cannot be deactivated].



- (b) Why the  $5\text{ V}$  voltage source couldn't give rise to a current? Can you draw any conclusions about the resistances of an ideal voltage and current source from this?



## Report

1. Fill up the theoretical parts of all the data tables.
2. Answers to the questions.
3. Discussion [brief summary of the experiment, key concepts learned, accuracy and consistency of the measured data with theory, difficulties experienced and your thoughts on those, and ways to improve]. Add pages if necessary.